

3.3 Backward Error Analysis for Hamiltonian Splitting Methods

As the relation

$$\mathcal{L}_H \phi = \{\phi, H\}$$

is linear in H , we may write, for Hamiltonians H_1, H_2 ,

$$\mathcal{L}_{H_1+H_2} = \mathcal{L}_{H_1} + \mathcal{L}_{H_2}.$$

The flow map of the system with Hamiltonian $H = H_1 + H_2$ is

$$\mathcal{F}_t = e^{t(\mathcal{L}_{H_1} + \mathcal{L}_{H_2})}.$$

On the other hand, the splitting method based on a composition of flows on H_1 and H_2 is

$$\mathcal{G}_h = e^{h\mathcal{L}_{H_1}} e^{h\mathcal{L}_{H_2}}.$$

The issue then becomes the comparison of the exponential series

$$e^{h(A+B)}$$

with

$$e^{hA} e^{hB},$$

for noncommuting operators A, B . Expanding these out using the exponential series we have

$$\begin{aligned} e^{h(A+B)} &= \text{Id} + h(A+B) + \frac{h^2}{2}(A+B)^2 + \frac{h^3}{6}(A+B)^3 + \mathcal{O}(h^4) \\ &= \text{Id} + h(A+B) + \frac{h^2}{2}(AB+BA+A^2+B^2) \\ &\quad + \frac{h^3}{6}(A^3+A^2B+AB^2+ABA+B^2A+BA^2+BAB+B^3) \\ &\quad + \mathcal{O}(h^4), \end{aligned}$$

whereas,

$$\begin{aligned}
 e^{hA} e^{hB} &= (\text{Id} + hA + \frac{h^2 A^2}{2} + \frac{h^3 A^3}{6} + \mathcal{O}(h^4)) \\
 &\quad \times (\text{Id} + hB + \frac{h^2 B^2}{2} + \frac{h^3 B^3}{6} + \mathcal{O}(h^4)) \\
 &= \text{Id} + h(A + B) + \frac{h^2}{2}(2AB + A^2 + B^2) \\
 &\quad + \frac{h^3}{6}(A^3 + B^3 + 3AB^2 + 3A^2B) + \mathcal{O}(h^4).
 \end{aligned}$$

So the difference is

$$\begin{aligned}
 e^{hA} e^{hB} - e^{h(A+B)} &= \frac{h^2}{2}(AB - BA) \\
 &\quad + \frac{h^3}{6}(2AB^2 + 2A^2B - BA^2 - BAB - B^2A - ABA) + \mathcal{O}(h^4) \\
 &= \frac{h^2}{2}[A, B] + \mathcal{O}(h^3),
 \end{aligned}$$

where $[A, B] = AB - BA$ is the commutator of A and B .

For a splitting method, we have, replacing A by $\mathcal{L}_{H_1}$ and B by $\mathcal{L}_{H_2}$,

$$e^{h\mathcal{L}_{H_1}} e^{h\mathcal{L}_{H_2}} - e^{h\mathcal{L}_H} = \frac{h^2}{2}[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}] + \mathcal{O}(h^3).$$

Observe that, for any real-valued functions f and H of phase space,

$$\mathcal{L}_H f = \nabla f^T \mathbf{J} \nabla H = \{f, H\}.$$

Hence

$$\mathcal{L}_{H_1} \mathcal{L}_{H_2} f = \{\{f, H_2\}, H_1\},$$

and, using this and skew-symmetry of the Poisson bracket,

$$[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}]f = \{\{f, H_2\}, H_1\} - \{\{f, H_1\}, H_2\} = \{\{f, H_2\}, H_1\} + \{\{H_1, f\}, H_2\}.$$

By the Jacobi identity,

$$\{\{f, H_2\}, H_1\} + \{\{H_2, H_1\}, f\} + \{\{H_1, f\}, H_2\} = 0.$$